

Dr. Jordi-Joan Giménez Gandia

Head of Technology, 5G-MAG Project Office

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PROFILE

Technology leader with 15+ years in media delivery across mobile and broadcast networks. I work in the Project Office of 5G-MAG – The Media Connectivity Association, managing the association's day-to-day activities and its technical programme. Track record of 30+ contributions to international standards (3GPP, DVB, ATSC), open-source programmes, EU research projects and research: 60+ papers, 1,000+ citations and 3 granted US patents.

EXPERIENCE

5G-MAG – The Media Connectivity Association · Geneva

Apr 2020 – present

Head of Technology, 5G-MAG Project Office

- Day-to-day management of the association: coordination of member projects and activities, work planning, logistics, the members' platform, website and social media.
- Organisation of workshops and events, such as the Future Media Townhall at IBC; member relations and communication with the media and international fora.
- Lead and maintainer of the 5G-MAG Reference Tools on GitHub, open implementations of 3GPP, MPEG, DVB and IETF specifications (2,300+ contributions in the last year; 430+ pull requests, 140+ reviews).
- 3GPP RAN1 delegate and feature lead for Rel-19 time-frequency interleaving (RRC parameters); testing and profiling with srsRAN and Open5GS, including NTN and MBS.

Institut für Rundfunktechnik (IRT) · Munich

Apr 2018 – Apr 2020

Project Lead & Research Engineer, Radio Systems Department

- 3GPP RAN1 delegate for LTE-based 5G Terrestrial Broadcast (Rel-16): contributions, simulations and coordination with 3GPP companies.
- Chair of the EBU 5G-Deployments industry group; delegate to the EBU Mobile Technology Standards group.
- Project leader for the H2020 5G-PPP projects 5G-Xcast and 5G-SOLUTIONS.
- MATLAB link- and network-level simulators for eMBMS Rel-14; software-defined-radio demonstrators (srsLTE, Amarisoft) for IBC 2018/2019 and MWC 2019; patents on RAN enhancements.

iTEAM Research Institute, Universitat Politècnica de València · Valencia

Feb 2011 – Mar 2018

Research Engineer (2011–2016) · Postdoctoral Researcher, APOSTD grant (2016–2018) · Mobile Communications Group

- Standardization of next-generation terrestrial TV technologies (DVB-T2, DVB-NGH, ATSC 3.0).
- Link-level simulation, propagation modelling, frequency planning and coverage estimation for broadcast and cellular networks.
- Drive tests and laboratory measurements, including an HEVC 120 fps pilot transmitter over DVB-T2.
- Industry projects with ETRI (ATSC 3.0), Panasonic and Teracom (WiB), BMW (DAB/eMBMS service continuity) and mobile operators.

Baa Consultors S.L.

Nov – Dec 2017

Technical expert, EU technical assistance for the implementation of digital TV in Colombia (EuropeAid).

Research stays: IRT, Munich (2015, 2017–2018) · Teracom AB, Stockholm (2013, 2014) · KTH Royal Institute of Technology, Stockholm (2014).

STANDARDIZATION · 30+ TECHNICAL CONTRIBUTIONS

- **3GPP RAN1** (2018 – present): feature lead for Rel-19 time-frequency interleaving; 8 contributions on LTE-based 5G Terrestrial Broadcast (Rel-16) with the EBU, BBC and IRT.
- **DVB** (2011 – 2018): 20 contributions to DVB-NGH, TM-MIMO and TM-WIB, including the TM-WIB final report.
- **ATSC 3.0**: proposals on Time Frequency Slicing and Advanced Network Planning to the TG3/S32-3 waveform group.
- **EBU** (2018 – 2020, at IRT): chair of the 5G-Deployments industry group; delegate to the Mobile Technology Standards group.

PROJECTS & CONTRACTS · 7 R&D PROJECTS, 8 INDUSTRY CONTRACTS

- **EU H2020:** 5G-SOLUTIONS (2019–20, €14.3M, IRT) · 5G-Xcast (2017–19, €8.0M, UPV).

- **IRT:** multicast delivery and HbbTV web apps in Android (2018, principal investigator).
- **National programmes:** extra-terrestrial digital broadcasting (Spanish Ministry of Economy, 2016–18) · non-uniform constellations for ATSC 3.0 (UPV, 2015–16) · FREEDOM (CDTI, 2012–13) · ENGINES (Celtic-Plus, 2010–).
- **Contracts:** ETRI, South Korea (TFS for cloud transmission, non-uniform constellations and LDM for ATSC 3.0, 2014–16) · EuropeAid, digital TV in Colombia (2017) · TV/LTE coexistence at 700 MHz in Peru and Argentina (2015–16) · Orange, Vodafone and Telefónica (2013) · SAPEC, Egatel and Axión (2012–13) · Abertis Retevisión, DVB-H/SH planning (2007–08).

Supervision: 2 M.Sc. theses (2016, 2018) and 2 final-degree projects (2012; 2013, 10/10 with honours), UPV.

PhD committees: Secretary of the examination committee for E. Iradier, "Power domain NOMA for 5G networks and beyond", UPV/EHU (2021).

EDUCATION

PhD in Telecommunication, UPV, 2015
Cum laude, International Doctorate mention

M.Sc. Technologies, Systems and Telecom Networks, UPV, 2011

M.Sc. Telecommunication Engineering, UPV, 2010

AWARDS & GRANTS

IBC Best Paper Award, 2016 (WiB)

IEEE BMSB Best Student Paper Award, 2017

Vodafone/ONO Best M.Sc. Thesis (DTT gap-fillers), COIT, 2012

Guest editor, IEEE Trans. Broadcasting special issues, 2022 and 2024

APOSTD postdoctoral grant (2016) · FPI-UPV predoctoral grant (2013)

PATENTS

US 11,916,655 B2 — Transmitter and/or receiver for transmitting and/or receiving radio information signals (IRT, 2024)

US 10,225,040 B2 — Broadcasting signal using combination of layered division multiplexing and channel bonding (ETRI, 2019)

US 9,917,666 B2 — Switching between DAB and eMBMS transmissions for a broadcast service (BMW, 2018)

US 2023/0053534 A1 — Cellular network broadcasting (IRT, application)

SELECTED PUBLICATIONS · full list at jordijoan.es and Google Scholar

- F. Gabin, D. Silhavy, T. Stockhammer, R. Bradbury, T. Lohmar, **J.J. Giménez**, J. Mika, "Advanced Media Delivery: Latest Updates on Streaming and Broadcast," IBC 2025.
- N. Dhollande, D. Butler, R. Bradbury, U. Rauschenbach, T. Biatek, T. Lohmar, **J.J. Giménez**, "Unlocking Network APIs for Media Production over Mobile Networks," IBC 2025.
- E. Garro et al., **J.J. Giménez**, D. Gómez-Barquero, "5G Mixed Mode: NR Multicast-Broadcast Services," IEEE Trans. Broadcasting, 2020.
- **J.J. Giménez** et al., "5G New Radio for Terrestrial Broadcast: A Forward-Looking Approach for NR-MBMS," IEEE Trans. Broadcasting, 2019.
- **J.J. Giménez**, D. Gómez-Barquero, J. Morgade, E. Stare, "Wideband Broadcasting: A Power-Efficient Approach to 5G Broadcasting," IEEE Communications Magazine, 2018.
- E. Stare, P. Klenner, **J.J. Giménez**, "WIB: A New System Concept for Digital Terrestrial Television," IBC 2016 — Best Paper Award.

SELECTED TALKS

Keynote, IEEE BMSB 2022 · Future Media Townhall, IBC 2026 · Fraunhofer FOKUS Media Web Symposium 2022, 2025, 2026 · FOSDEM 2024 · invited talk, workshop on 5G in content production and distribution (2024) · UWA Summit 2023 · 5G-MAG at IBC 2023 · 5G-Xcast tutorial, IEEE BMSB 2019

SKILLS

Leadership: association management, programme and project management, member and stakeholder relations, events and workshops, communication and outreach, technology strategy, standards delegation and consensus building, open-source community building, EU proposals (H2020, Horizon Europe), moderation and public speaking.

Technology: 3GPP RAN1, 5G multicast/broadcast (MBS), 5G Media Streaming, network APIs (CAMARA), NTN, MPEG-I/V3C and XR, DVB and ATSC 3.0, network planning and propagation, PHY and signal processing.

Tools: srsRAN, Open5GS, Amarisoft, 5G-MAG Reference Tools, MATLAB, Atoll, ICS Telecom, IRT FRANSY, GitHub.

Languages: Catalan and Spanish (native), English (professional), German and French (basic).